COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	**CPE212 / Automating Server Management**	**SECTION:**	**CPE31S2**
DATE PERFORMED:	**Jul 30 2026**	**DATE SUBMITTED:**	**Jul 30 2026**
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ACTIVITY NO. 2 **SSH Key-Based Authentication and Setting up Git**

1. PROCEDURE OUTPUT

Task 1: Create an SSH Key Pair for User Authentication

1. The simplest way to generate a key pair is to run ssh-keygen without arguments. In this case, it will prompt for the file in which to store keys. First, the tool asked where to save the file. SSH keys for user authentication are usually stored in the users .ssh directory under the home directory. However, in enterprise environments, the location is often different. The default key file name depends on the algorithm, in this case id_rsa when using the default RSA algorithm. It could also be, for example, id_dsa or id_ecdsa.
2. Issue the command ssh-keygen -t rsa -b 4096. The algorithm is selected using the -t option and key size using the -b option.
3. When asked for a passphrase, just press enter. The passphrase is used for encrypting the key, so that it cannot be used even if someone obtains the private key file. The passphrase should be cryptographically strong.
4. Verify that you have created the key by issuing the command ls -la .ssh. The command should show the .ssh directory containing a pair of keys. For example, id_rsa.pub and id_rsa.

```
kenzo@workstation:~$ ssh-keygen -t rsa -b 4096
Generating public/private rsa key pair.
'Enter file in which to save the key (/home/kenzo/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in '
Your public key has been saved in '.pub
The key fingerprint is:
SHA256:SOs4iT2t+Kbo4QV9Kgb3b8xvmnWh0qFHvAuFci7Q83w kenzo@workstation
The key's randomart image is:
+---[RSA 4096]----+
|                 |
|                 |
|       .         |
|  .. . =         |
|..o.+.= S .      |
|...=o% = + .     |
| +.oOoX E .      |
|o.+..*+B.o       |
|ooo+o.++o        |
+----[SHA256]-----+
```

```
kenzo@workstation:~$ ls -a .ssh
.  ..  known_hosts  known_hosts.old
```

Task 2: Copying the Public Key to the remote servers

1. To use public key authentication, the public key must be copied to a server and installed in an authorized_keys file. This can be conveniently done using the ssh-copy-id tool.

2. Issue the command similar to this: ssh-copy-id -i ~/.ssh/id_rsa user@host

```
kenzo@workstation:~$ ssh-copy-id kenzo@192.168.56.7
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to fi
lter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are p
rompted now it is to install the new keys
kenzo@192.168.56.7's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'kenzo@192.168.56.7'"
and check to make sure that only the key(s) you wanted were added.
```

```
kenzo@workstation:~$ ssh-copy-id kenzo@192.168.56.8
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to fi
lter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are p
rompted now it is to install the new keys
kenzo@192.168.56.8's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'kenzo@192.168.56.8'"
and check to make sure that only the key(s) you wanted were added.
```

3. Once the public key has been configured on the server, the server will allow any connecting user that has the private key to log in. During the login process, the client proves possession of the private key by digitally signing the key exchange.

4. On the local machine, verify that you can SSH with Server 1 and Server 2. What did you notice? Did the connection ask for a password? If not, why

```
kenzo@workstation:~$ ssh kenzo@server1
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-134-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check
your Internet connection or proxy settings

Last login: Thu Jul 30 08:54:13 2026 from 192.168.56.6
kenzo@server1:~$
```

```
kenzo@workstation:~$ ssh kenzo@server2
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-134-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check
your Internet connection or proxy settings

Last login: Thu Jul 16 09:43:55 2026 from 192.168.56.6
```

It did not ask for a password. It is most probably because I copied the ssh key into my servers, which then the server checks its authorized_keys to find the public key.

Reflections:
Answer the following:
1. How will you describe the ssh-program? What does it do?

Basically the ssh or secure shell. is a tool that is used to securely access and manage a computer remotely. It is able to create a secure and encrypted session between the local computer and remote machine.

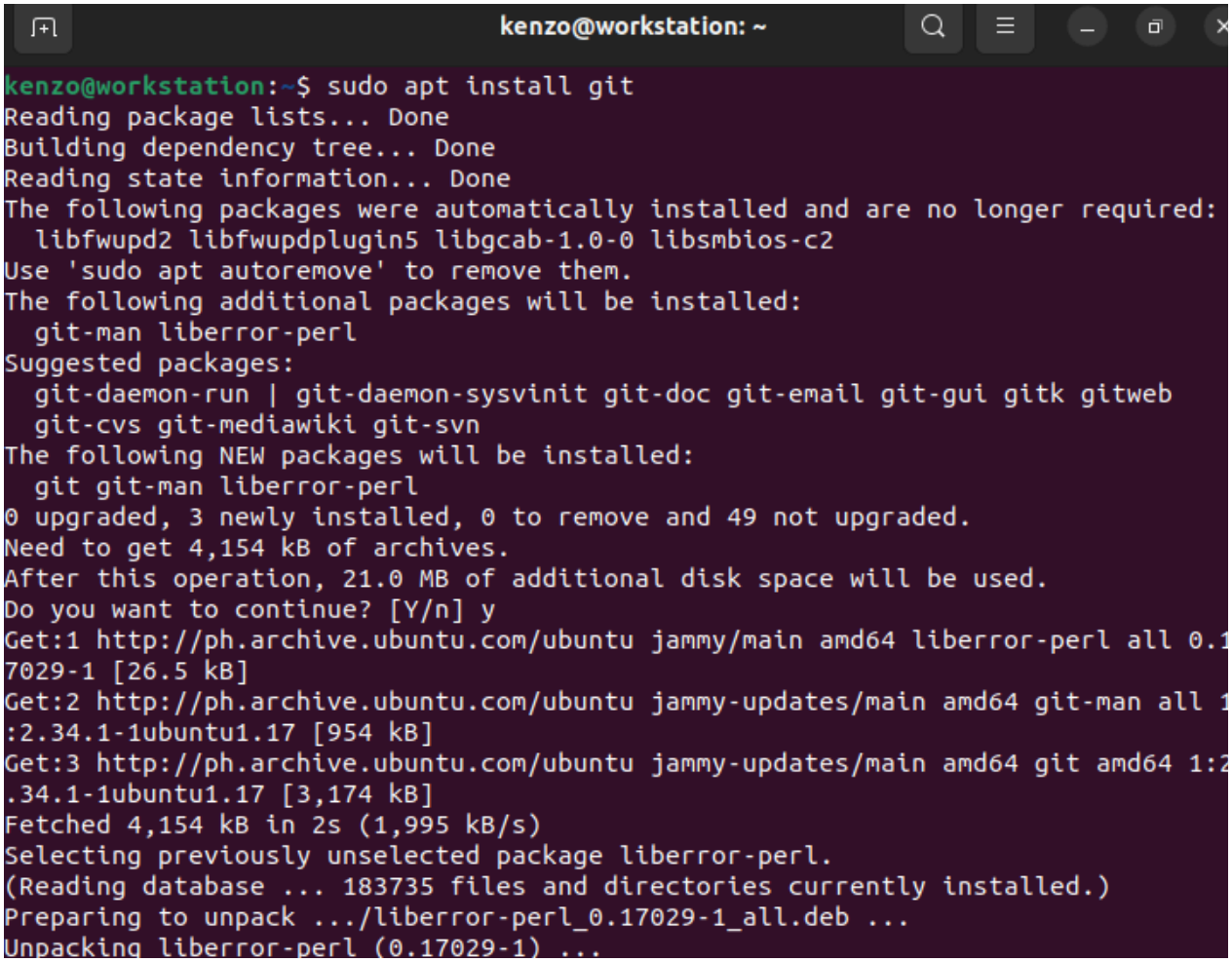
2. How do you know that you already installed the public key to the remote servers?

We could verify by logging in to the remote server and checking the ~/.ssh/authorized keys directory.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

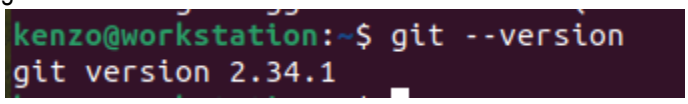
Part 2: Discussion
Provide screenshots for each task.
It is assumed that you are done with the last activity (Activity 2: SSH Key-Based Authentication).
Set up Git
At the heart of GitHub is an open-source version control system (VCS) called Git. Git is responsible for everything GitHub-related that happens locally on your computer. To use Git on the command line, you'll need to download, install, and configure Git on your computer. You can also install GitHub CLI to use GitHub from the command line. If you don't need to work with files locally, GitHub lets you complete many Git-related actions directly in the browser, including:
• Creating a repository
• Forking a repository
• Managing files
• Being social

1. On the local machine, verify the version of your git using the command which git. If a directory of git is displayed, then you don't need to install git. Otherwise, to install git, use the following command: sudo apt install git



2. After the installation, issue the command which git again. The directory of git is usually installed in this location: user/bin/git.

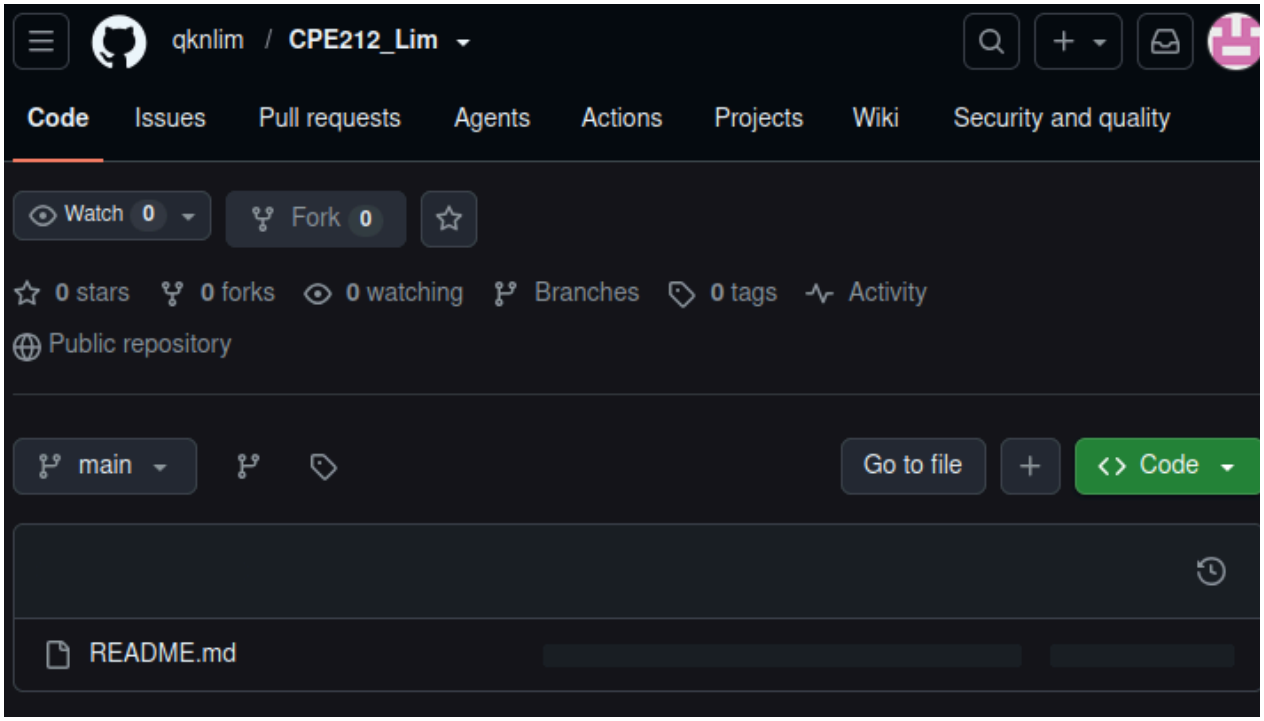
3. The version of git installed in your device is the latest. Try issuing the command git --version to know the version installed.



4. Using the browser in the local machine, go to www.github.com.

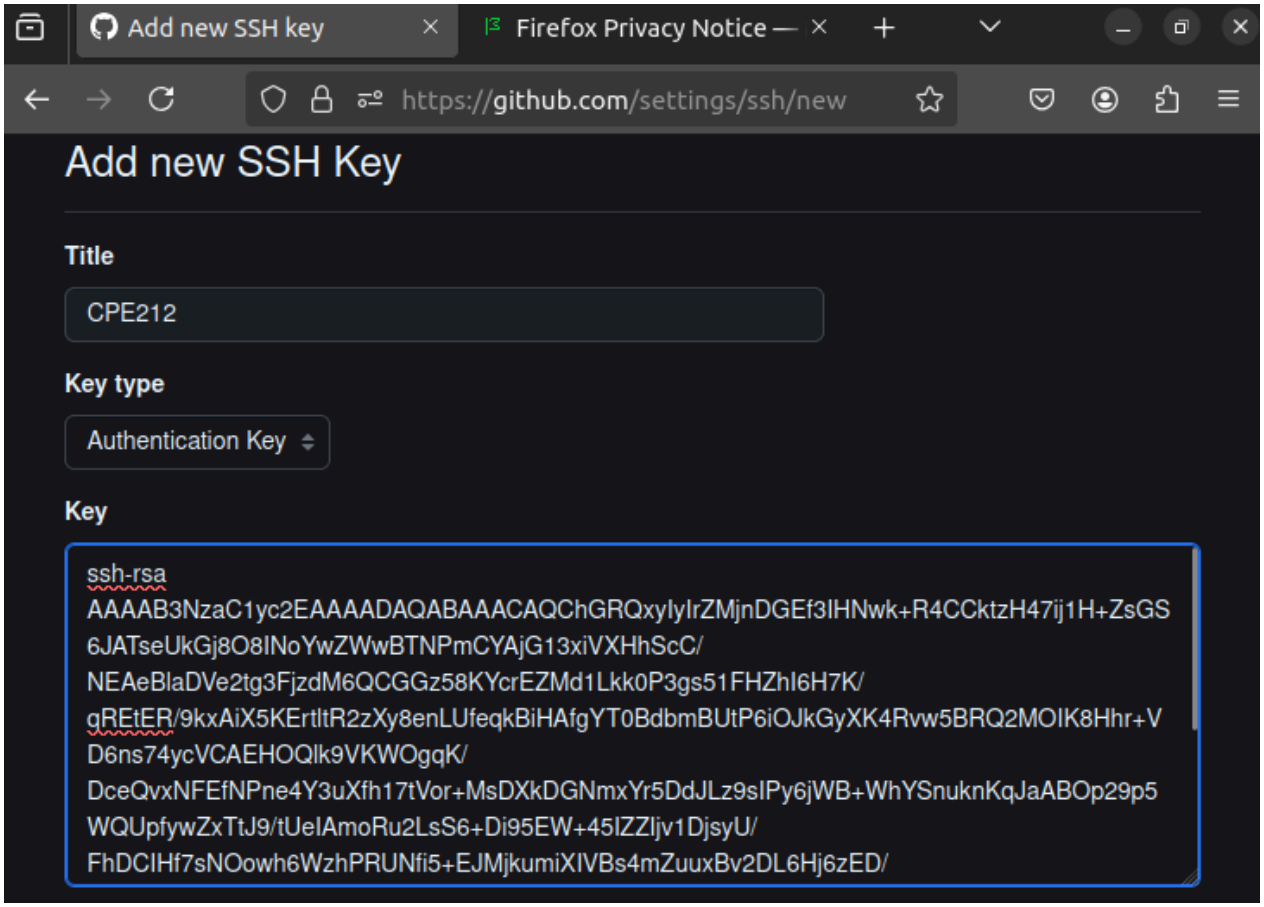
5. Sign up in case you don't have an account yet. Otherwise, login to your GitHub Account.

a. Create a new repository and name it as CPE232_yourname. Check Add a README file and click Create repository.

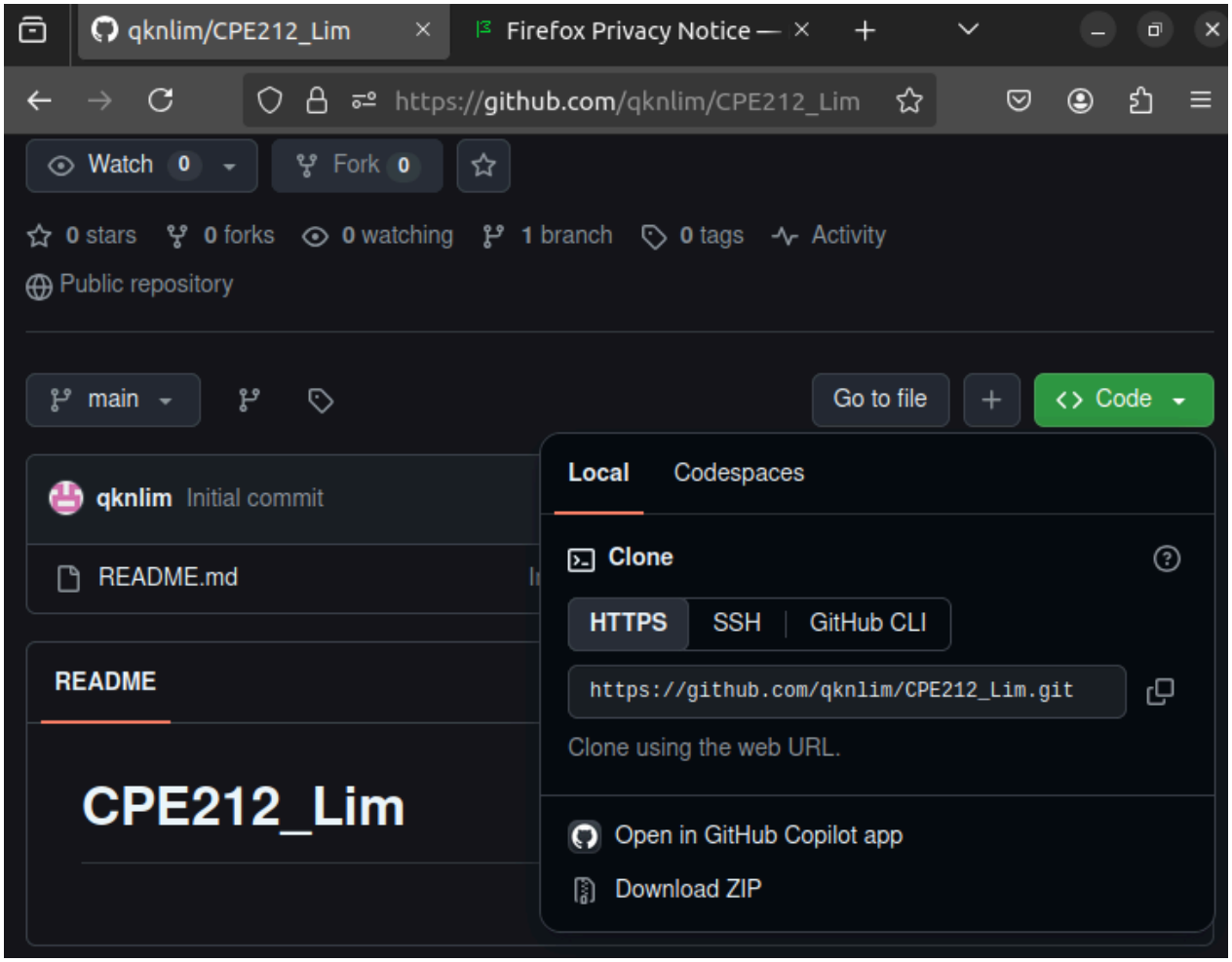


b. Create a new SSH key on GitHub. Go your profile's setting and click SSH and GPG keys. If there is an existing key, make sure to delete it. To create a new SSH keys, click New SSH Key. Write CPE232 key as the title of the key.

c. On the local machine's terminal, issue the command cat .ssh/id_rsa.pub and copy the public key. Paste it on the GitHub key and press Add SSH Key.



d. Clone the repository that you created. In doing this, you need to get the link from GitHub. Browse to your repository as shown below. Click on the Code drop down menu. Select SSH and copy the link.



e. Issue the command git clone followed by the copied link. For example, git clone git@github.com:jvtaylar-cpe/CPE232_yourname.git. When prompted to continue connecting, type yes and press enter.

```
kenzo@workstation:~$ git clone git@github.com:qknlim/CPE212_Lim.git
Cloning into 'CPE212_Lim'...
The authenticity of host 'github.com (20.205.243.166)' can't be established.
ED25519 key fingerprint is SHA256:+DiY3wvvV6TuJJhbpZisF/zLDA0zPMSvHdkr4UvCOqU
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])?
```

f. To verify that you have cloned the GitHub repository, issue the command ls. Observe that you have the CPE232_yourname in the list of your directories. Use CD command to go to that directory and LS command to see the file README.md.

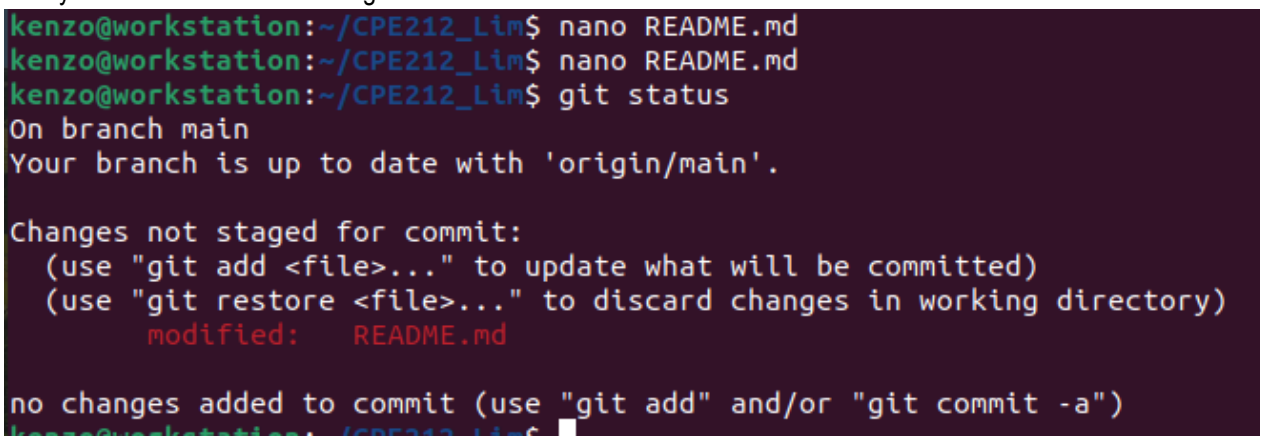
```
kenzo@workstation:~/CPE212_Lim$ ls
README.md
```

g. Use the following commands to personalize your git.
• git config --global user.name “Your Name”
• git config --global user.email yourname@email.com
• Verify that you have personalized the config file using the command cat ~/.gitconfig



h. Edit the README.md file using nano command. Provide any information on the markdown file pertaining to the repository you created. Make sure to write out or save the file and exit.

i. Use the git status command to display the state of the working directory and the staging area. This command shows which changes have been staged, which haven’t, and which files aren’t being tracked by Git. Status output does not show any information regarding the committed project history. What is the result of issuing this command?



j. Use the command git add README.md to add the file into the staging Area.

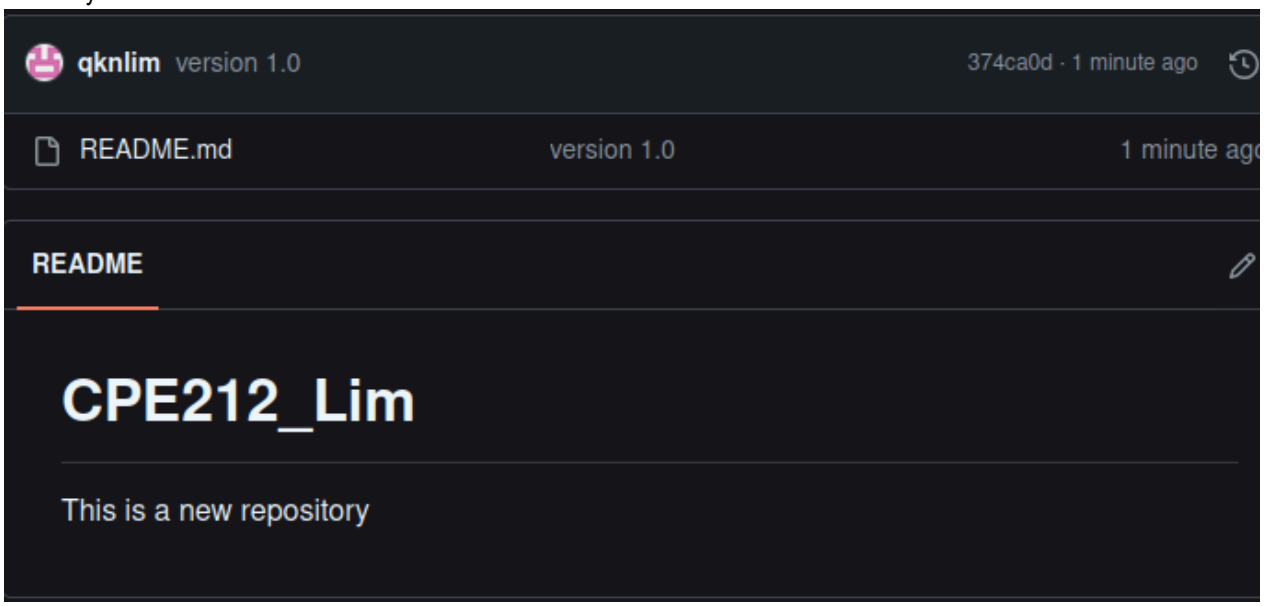
k. Use the git commit -m “your message” to create a snapshot of the staged changes along the timeline of the Git projects history. The use of this command is required to select the changes that will be staged for the next Commit.

```
no changes added to commit (use "git add" and/or "git commit -a")
kenzo@workstation:~/CPE212_Lim$ git add README.md
kenzo@workstation:~/CPE212_Lim$ git commit -m "version 1.0"
[main 374ca0d] version 1.0
 1 file changed, 2 insertions(+), 1 deletion(-)
```

l. Use the command git push <remote><branch> to upload the local repository content to GitHub repository. Pushing means to transfer commits from the local repository to the remote repository. As an example, you may issue git push origin main.

```
kenzo@workstation:~/CPE212_Lim$ git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Writing objects: 100% (3/3), 278 bytes | 278.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:qknlim/CPE212_Lim.git
   300a8e4..374ca0d  main -> main
kenzo@workstation:~/CPE212_Lim$
```

m. On the GitHub repository, verify that the changes have been made to README.md by refreshing the page. Describe the README.md file. You can notice the how long was the last commit. It should be some minutes ago and the message you typed on the git commit command should be there. Also, the README.md file should have been edited according to the text you wrote.



3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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